

Risultati

ANALISI DESCRITTIVE DELLE VARIABILI

- 1 variabile numerica
- 9 variabili binarie
- 2 variabili categoriche
- 1 variabile ordinale

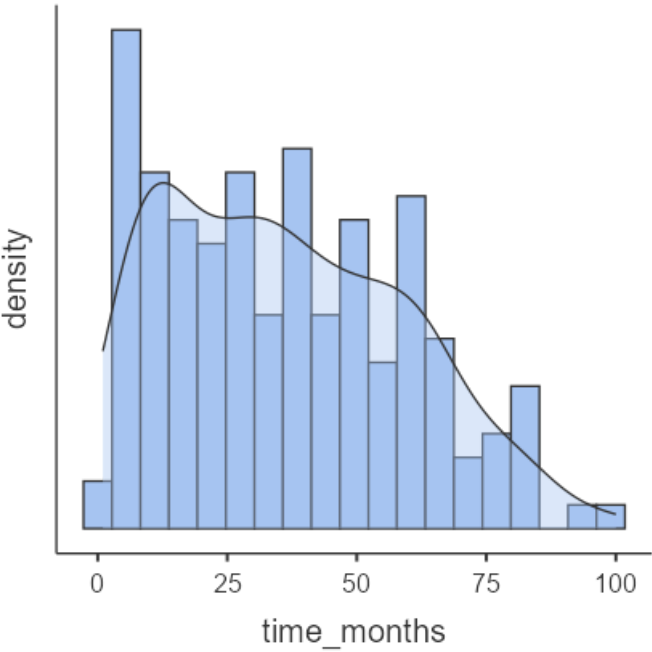
Descrittive variabile numerica

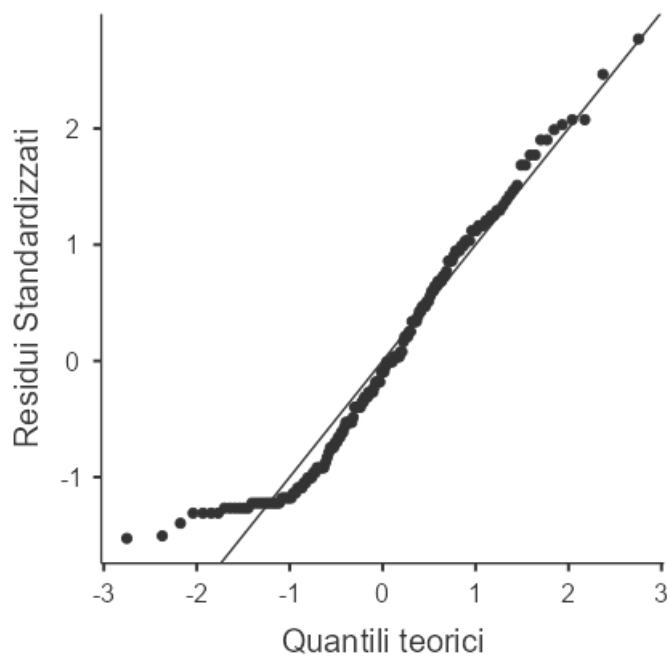
Descrittive

	N	Media	Mediana	SD	Minimo	Massimo
time_months	169	36.2	34.0	23.0	1.00	100

Grafici

time_months





Univariate Data Analysis (Categorical Data): Binarie

Frequencies distribution

Frequencies and Odds of sex

Levels	Counts	Cumulative Counts	Proportions	Cumulative Proportions	% of Total	Cumulative %	Odds
0	110	110	0.651	0.651	65.1%	65.1%	1.864
1	59	169	0.349	1.000	34.9%	100.0%	0.536
Missings	0	169	0.000	1.000	0.0%	100.0%	0.000

0 = maschio

1 = femmina

Frequencies and Odds of stage

Levels	Counts	Cumulative Counts	Proportions	Cumulative Proportions	% of Total	Cumulative %	Odds
0	53	53	0.314	0.314	31.4%	31.4%	0.457
1	116	169	0.686	1.000	68.6%	100.0%	2.189
Missings	0	169	0.000	1.000	0.0%	100.0%	0.000

0 = tumore in stadio iniziale (stadio 0 o I)

1 = tumore in stadio avanzato (stadio II, III o IV)

Frequencies and Odds of risk

Levels	Counts	Cumulative Counts	Proportions	Cumulative Proportions	% of Total	Cumulative %	Odds
0	32	32	0.189	0.189	18.9%	18.9%	0.234
1	137	169	0.811	1.000	81.1%	100.0%	4.281
Missings	0	169	0.000	1.000	0.0%	100.0%	0.000

0 = intermedio

1 = alto

Frequencies and Odds of autologous_stem_cell_transplantation

Levels	Counts	Cumulative Counts	Proportions	Cumulative Proportions	% of Total	Cumulative %	Odds
0	155	155	0.9172	0.917	91.7%	91.7%	11.0714
1	14	169	0.0828	1.000	8.3%	100.0%	0.0903
Missings	0	169	0.0000	1.000	0.0%	100.0%	0.0000

0 = no

1 = si

Frequencies and Odds of radiation

Levels	Counts	Cumulative Counts	Proportions	Cumulative Proportions	% of Total	Cumulative %	Odds
0	60	60	0.355	0.355	35.5%	35.5%	0.550
1	109	169	0.645	1.000	64.5%	100.0%	1.817
Missings	0	169	0.000	1.000	0.0%	100.0%	0.000

0 = no

1 = si

Frequencies and Odds of UH_or_FH

Levels	Counts	Cumulative Counts	Proportions	Cumulative Proportions	% of Total	Cumulative %	Odds
0	136	136	0.805	0.805	80.5%	80.5%	4.121
1	33	169	0.195	1.000	19.5%	100.0%	0.243
Missings	0	169	0.000	1.000	0.0%	100.0%	0.000

0 = UH

1 = FH

Frequencies and Odds of MYCN_status

Levels	Counts	Cumulative Counts	Proportions	Cumulative Proportions	% of Total	Cumulative %	Odds
0	45	45	0.266	0.266	26.6%	26.6%	0.363
1	124	169	0.734	1.000	73.4%	100.0%	2.756
Missings	0	169	0.000	1.000	0.0%	100.0%	0.000

0 = amplificazione
1 = non amplificazione

Frequencies and Odds of surgical_methods

Levels	Counts	Cumulative Counts	Proportions	Cumulative Proportions	% of Total	Cumulative %	Odds
0	111	111	0.657	0.657	65.7%	65.7%	1.914
1	58	169	0.343	1.000	34.3%	100.0%	0.523
Missings	0	169	0.000	1.000	0.0%	100.0%	0.000

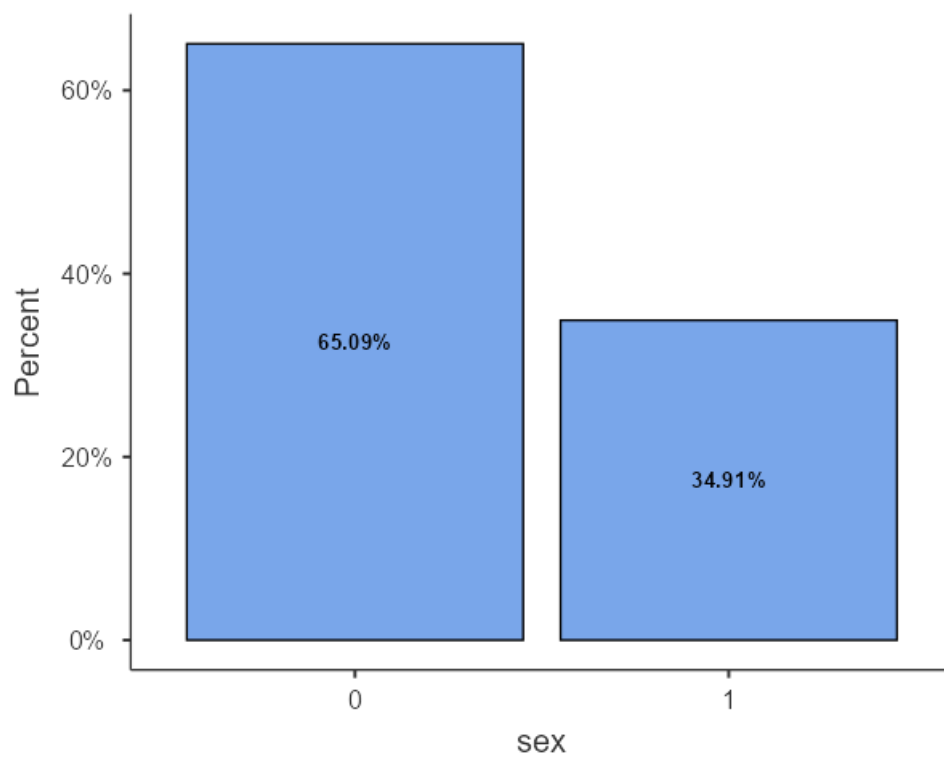
0 = resezione totale lorda: Tutto il tumore visibile è stato rimosso
1 = resezione subtotale: Solo una parte del tumore è stata rimossa

Frequencies and Odds of outcome

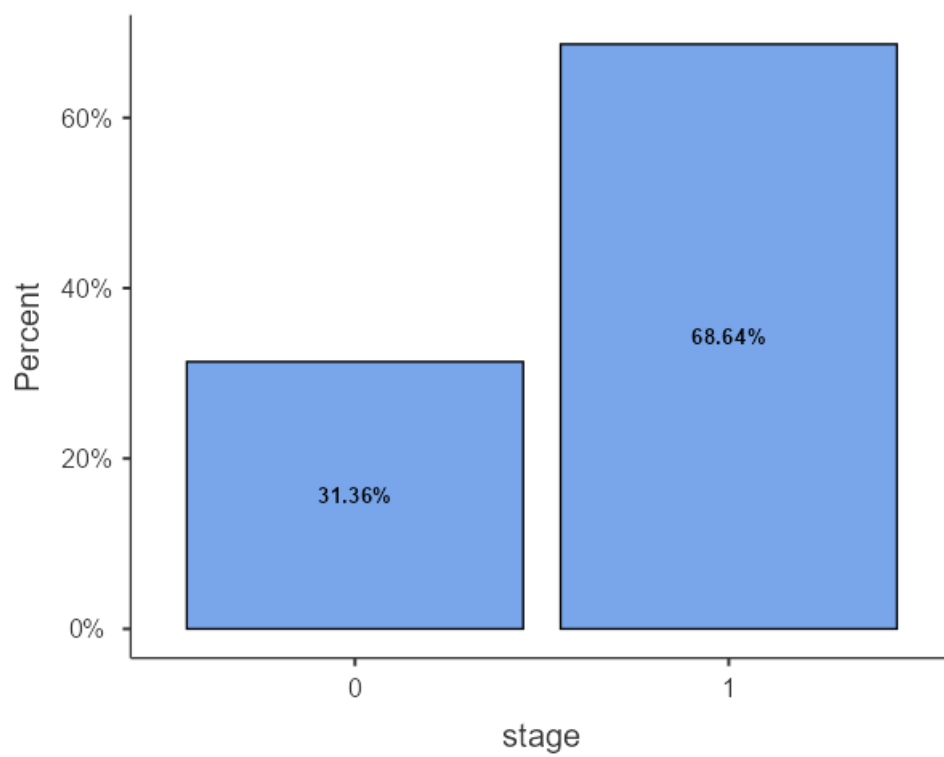
Levels	Counts	Cumulative Counts	Proportions	Cumulative Proportions	% of Total	Cumulative %	Odds
0	92	92	0.544	0.544	54.4%	54.4%	1.195
1	77	169	0.456	1.000	45.6%	100.0%	0.837
Missings	0	169	0.000	1.000	0.0%	100.0%	0.000

0 = sopravvivenza
1 = morte

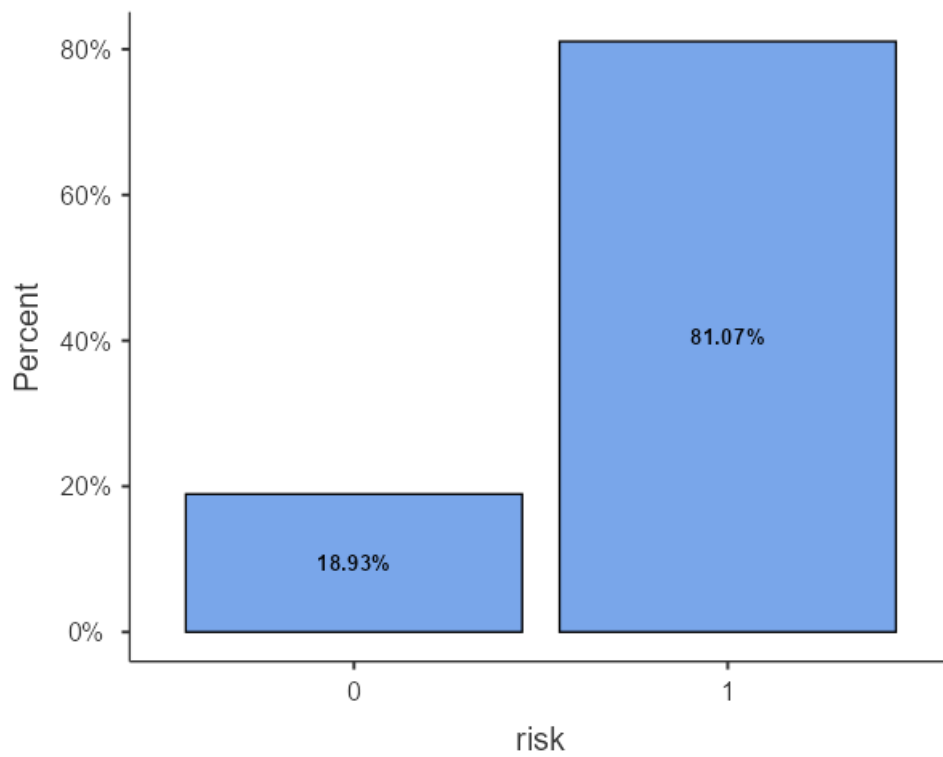
Bar Plot sex



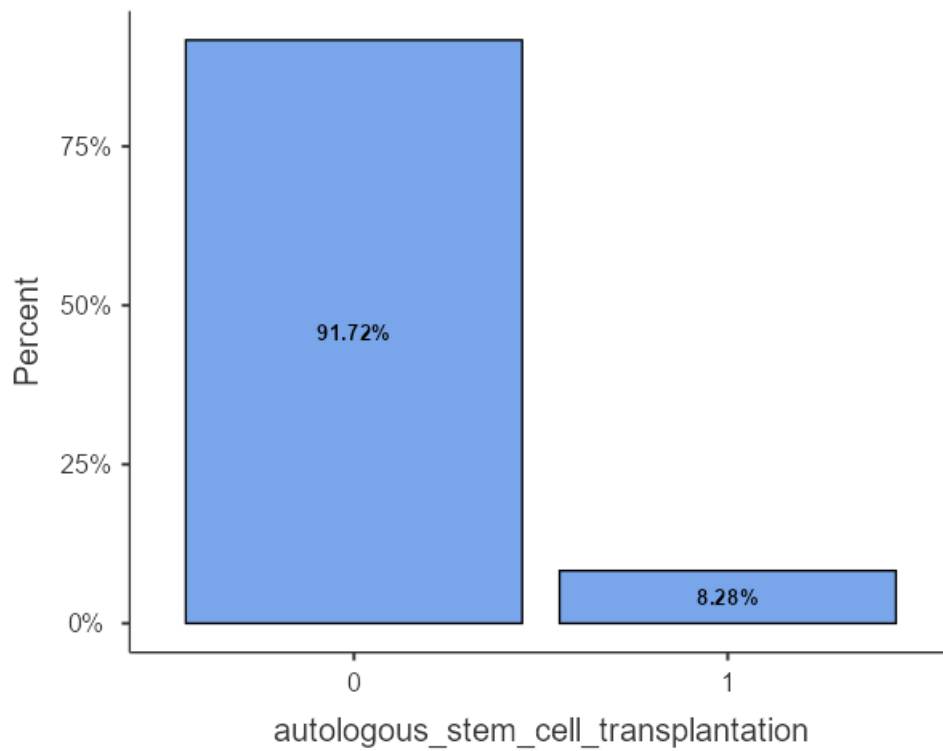
Bar Plot stage



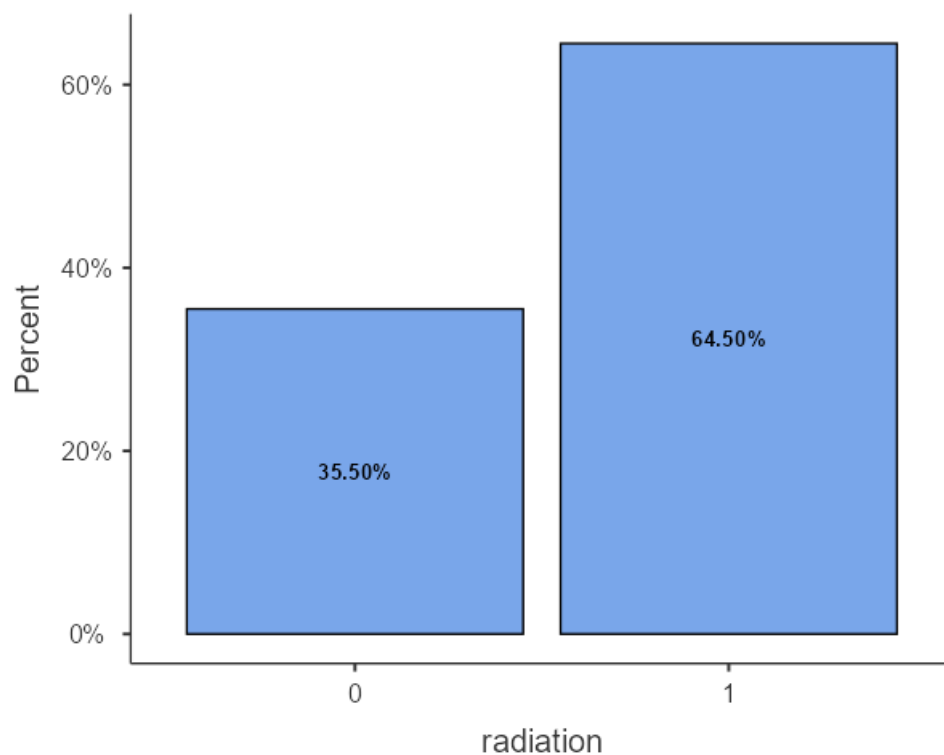
Bar Plot



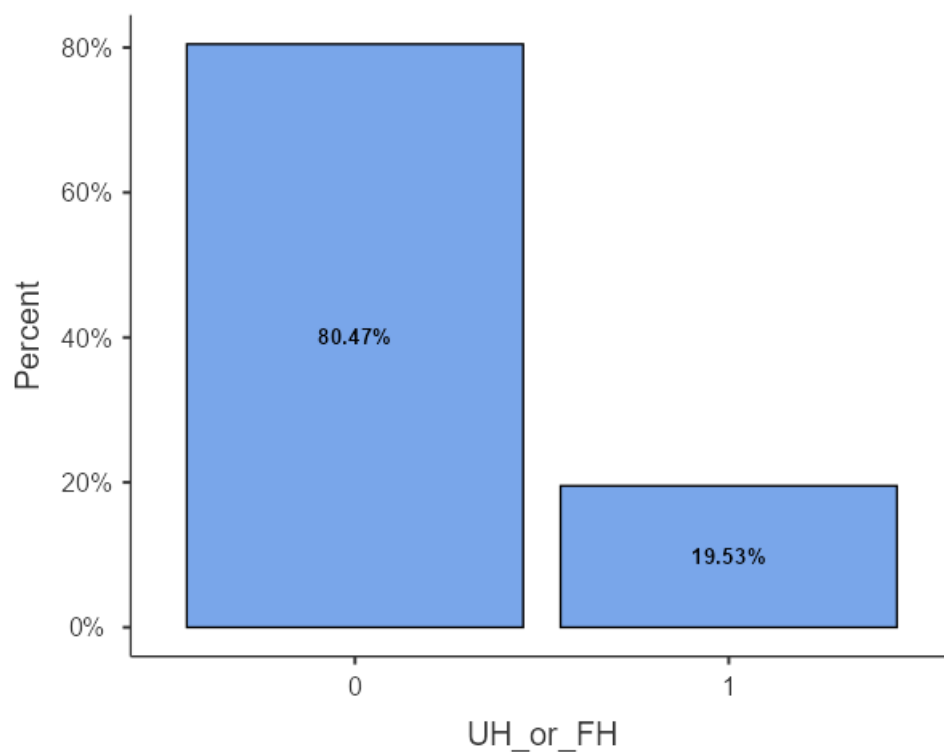
Bar Plot autologous_stem_cell_transplantation



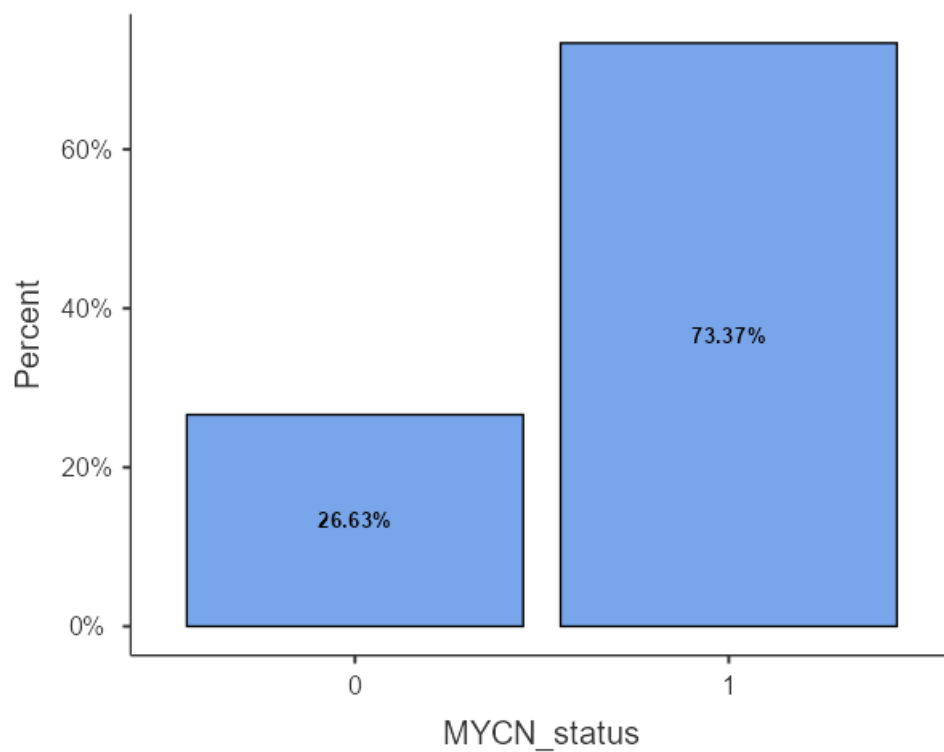
Bar Plot radiation



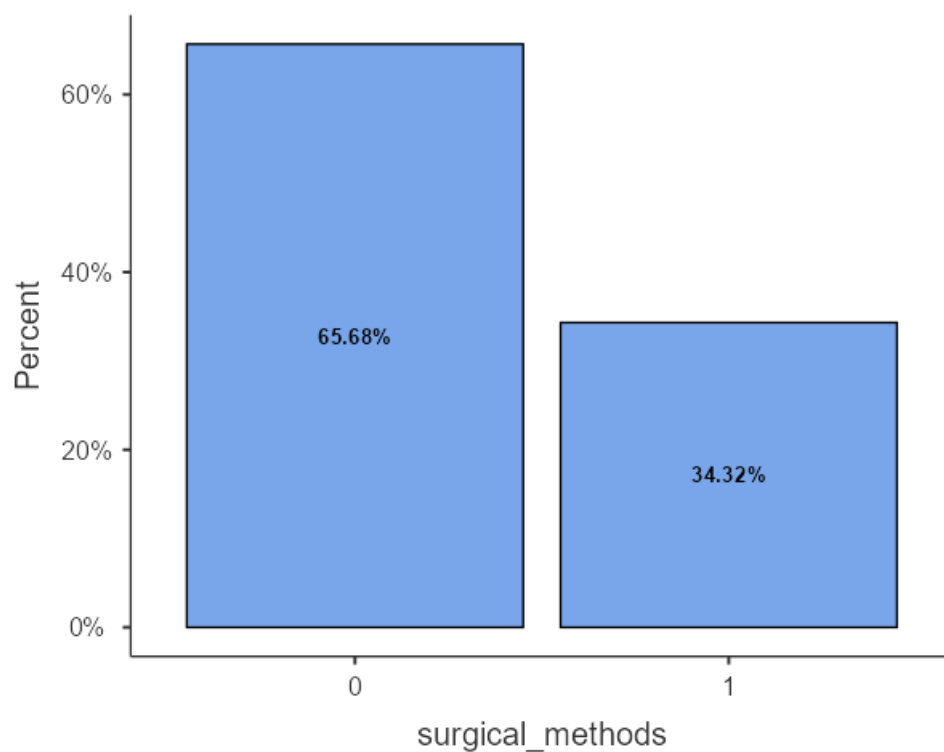
Bar Plot UH_or_FH



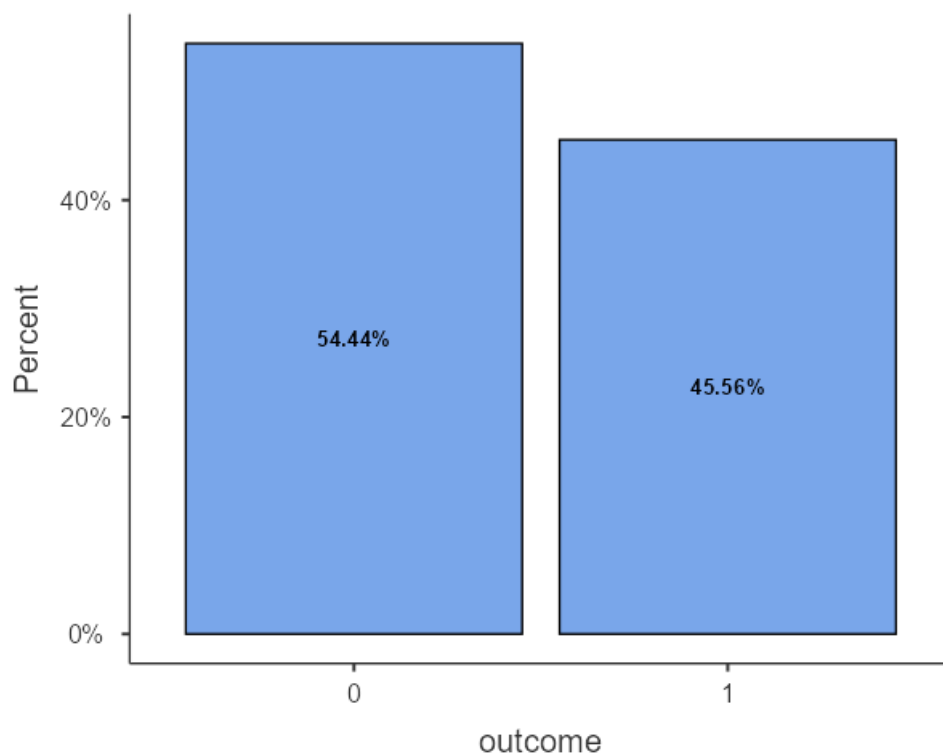
Bar Plot MYCN_status



Bar Plot surgical_methods



Bar Plot outcome



Univariate Data Analysis (Categorical Data)

Frequencies distribution

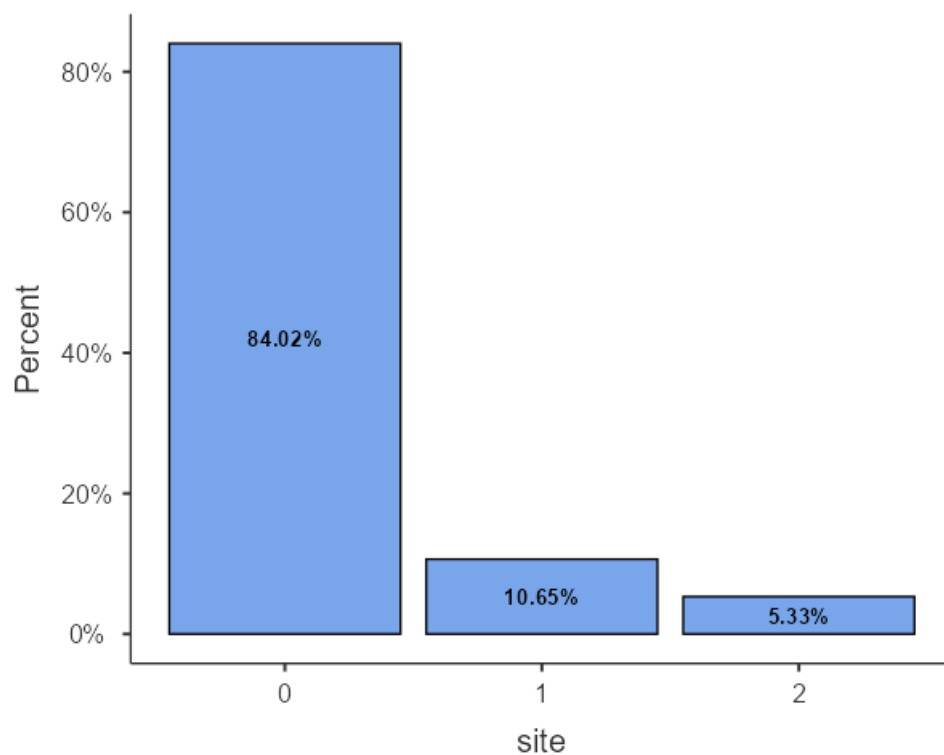
Frequencies and Odds of site

Levels	Counts	Cumulative Counts	Proportions	Cumulative Proportions	% of Total	Cumulative %	Odds
0	142	142	0.8402	0.840	84.0%	84.0%	5.2593
1	18	160	0.1065	0.947	10.7%	94.7%	0.1192
2	9	169	0.0533	1.000	5.3%	100.0%	0.0563
Missings	0	169	0.0000	1.000	0.0%	100.0%	0.0000

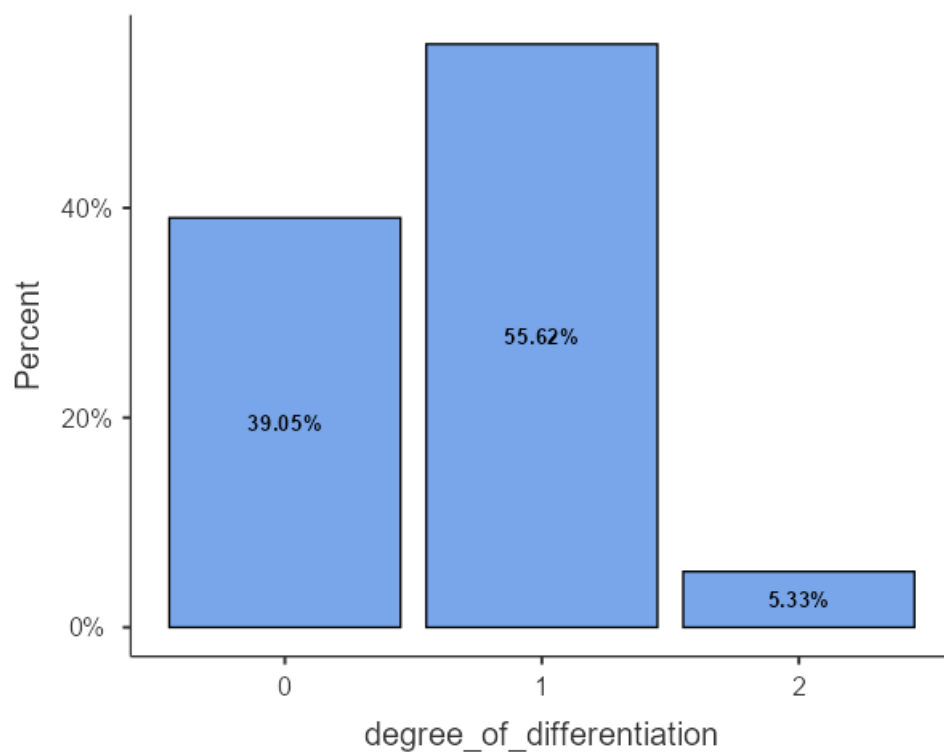
Frequencies and Odds of degree_of_differentiation

Levels	Counts	Cumulative Counts	Proportions	Cumulative Proportions	% of Total	Cumulative %	Odds
0	66	66	0.3905	0.391	39.1%	39.1%	0.6408
1	94	160	0.5562	0.947	55.6%	94.7%	1.2533
2	9	169	0.0533	1.000	5.3%	100.0%	0.0563
Missings	0	169	0.0000	1.000	0.0%	100.0%	0.0000

Bar Plot site



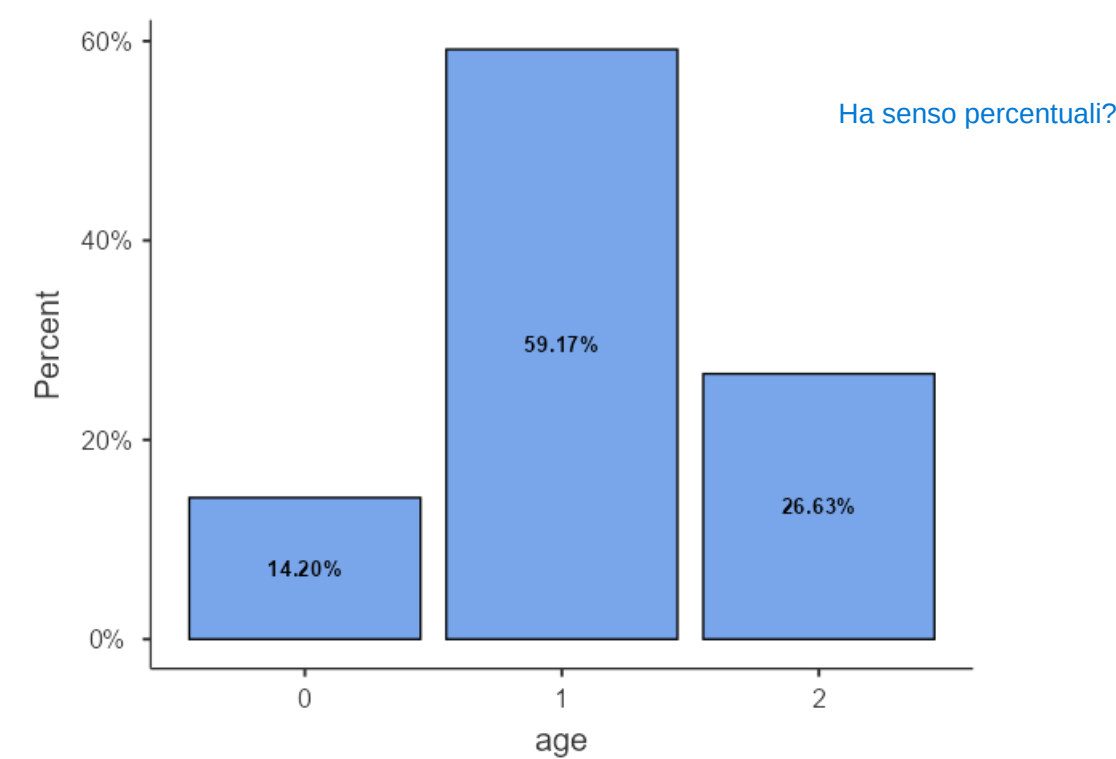
Bar Plot degree_of_differentiation



Descrittive variabile ordinale age

Descrittive	
	age
N	169
Media	1.12
Mediana	1
Moda	1.00
Deviazione standard	0.629

Bar Plot age



DOMANDA SCIENTIFICA A CUI RISPONDERE UTILIZZANDO I DATI:
C'è una associazione/correlazione tra l'amplificazione del gene MYCN e il rischio clinico o lo stadio del neuroblastoma nei pazienti con neuroblastoma?
Utilizzo: chi-quadro

Tabelle di Contingenza (MYCN vs risk)

Tabelle di Contingenza			
	MYCN_status		
risk	0	1	Totale
0	0	32	32
1	45	92	137
Totale	45	124	169

Le persone con rischio intermedio hanno il gene MYCN tutti non amplificato. Invece, chi ha un rischio alto ha, per la maggior parte, l'ampliamento del gene MYCN.

Test χ^2

	Valore	gdl	p
χ^2	14.3	1	<.001
N	169		

p piccolo (minore di 0,005): altamente significativo

Misure Comparative

	Valore	95% Intervallo di fiducia	
		Inferiore	Superiore
Log Odds-Ratio	-3.46 ^a	-6.28	-0.649
Odds-Ratio	0.0313 ^a	0.00187	0.522

^a Applicata la Correzione di Haldane-Anscombe

[3]

OR molto inferiore a 1 quindi chi è a basso rischio ha molte meno probabilità di avere MYCN amplificato rispetto a chi è ad alto rischio.

L'intervallo non include 1: significativo

Tabelle di Contingenza (MYCN vs stage)

Tabelle di Contingenza

stage	MYCN_status		Totale
	0	1	
0	9	44	53
1	36	80	116
Totale	45	124	169

Le persone con neuroblastoma allo stadio iniziale ancora non hanno in molti il gene MYCN amplificato (9). Invece, quando lo stadio diventa più avanzato più persone hanno il gene amplificato ma la maggior parte non ce l'ha amplificato anche in questo caso.

Test χ^2

	Valore	gdl	p
χ^2	3.68	1	0.055
N	169		

Il $p > 0,005$ ci dice che il test non raggiunge la significatività statistica. C'è associazione tra le 2 variabili ma non abbastanza se usiamo un intervallo di confidenza del 95%.

Misure Comparative

	Valore	95% Intervallo di fiducia	
		Inferiore	Superiore
Log Odds-Ratio	-0.788	-1.61	0.0294
Odds-Ratio	0.455	0.201	1.03

L'odds è comunque minore di 1 quindi i pazienti in stadio avanzato hanno meno probabilità di avere il gene non amplificato ma l'intervallo di confidenza include 1 (non significativo).

Riferimenti

[1] The jamovi project (2024). *jamovi*. (Version 2.6) [Computer Software]. Retrieved from <https://www.jamovi.org>.

[2] R Core Team (2024). *R: A Language and environment for statistical computing*. (Version 4.4) [Computer software]. Retrieved from <https://cran.r-project.org>. (R packages retrieved from CRAN snapshot 2024-08-07).

[3] Meyer, D., Zeileis, A., Hornik, K., Gerber, F., & Friendly, M. (2023). *vcd: Visualizing Categorical Data*. [R package]. Retrieved from <https://cran.r-project.org/package=vcd>.